1. GENERAL INSTRUCTIONS:

The following instructions apply to boiler work proper, work done by welding, either oxy-acetylene or arc welding, is covered by Maintenance Regulations No 7 and No 8

The extent of boiler work to be done in back shops is governed by the classification of repairs, see Maintenance Regulation No 63 and the boiler report Form 539 sent to shop with each individual engine

The standard method of doing each particular operation will be shown herein.

2. REMOVAL OF TUBES:

Chip off the beads at the back tube sheet, and when this is being done, carefully chip the tube sheet bridges to leave a smooth surface unless tube sheet is reported to be renewed

Tubes to be cut off inside the front tube sheet, with the standard mechanical tube cutter.

Tubes to be removed from boiler, care being taken to protect the hole through which the tubes are passed out.

Remove tube ends from front tube sheet, taking care that tube holes are not damaged.

Thoroughly clean, chip and scale interior of boiler, preparatory to examination as called for by Board of Railway Commissioners General Order No 473

3. PREPARATION OF TUBE SHEETS FOR SETTING TUBES:

Tube sheets must be properly straightened before tube holes are drilled and before tubes are applied Holes in new back tube sheets to be drilled one-sixteenth of an inch smaller than outside diameter of the tube Tube holes must be smooth and edges of holes chamfered to a radius of one-sixteenth of an inch on both sides of front and back tube sheets. In old back tube sheets all tube holes to be reamed with straight reamer to insure their being perfectly round and of uniform diameter If holes, in either tube sheet, are one-eighth of an inch or more larger than the outside diameter of tubes the tube sheet must be renewed

When new tube sheets are applied particular care must be taken to insure that the fire-box and front tube sheets are parallel, so that the tube lengths throughout the boiler will be equal.

4. PREPARATION OF TUBES FOR SETTING:

Tubes must be straight and back end of tubes must be so swedged that they can be driven into the copper ferrule by striking three or four sharp blows with a four-pound hammer Both ends of tubes must be properly annealed Ends of tubes both new and safe ended, must be cut off square and be free from burrs and scale